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Department of Chemistry
Jahangirnangr University
3rd Year B.Sc. (Hon's) Final Examination 2024
Course Code: CHEM 340; Course Title: Chemical Spectroscopy
Full Marks: 70; Time: 4 hours

Answer any five (05) of the following Question

1. (a) Define molecular spectroscopy. Classify spectroscopy based on the nature of interaction between atoms / molecules and radiation. Which of these require the least and the highest energy for transitions? 3
   (b) List the main forms of molecular energies. Explain whether molecular energies are continuous or quantized, giving proper justification. 4
   (c) What are the principal factors affecting the intensity of emission spectral lines? Explain each briefly. 4
   (d) Draw the schematic of a fluorescence spectrometer and mention the role of the monochromators used in it. 3

2. (a) State the criteria for a molecule to be microwave active. Provide examples of at least four molecules that can exhibit a rotational microwave absorption spectrum. 1+2
   (b) Discuss the limitations of microwave spectroscopy. 2
   (c) In the rotational spectrum of CO at 25 °C, the J = 3 → J = 4 transition is more intense than the J = 0 → J = 1 transition. Explain. 3
   (d) Explain the reason why the rotational constants of $^{12}C^{16}O$ and $^{13}C^{16}O$ are 1.92118 cm^{-1} and 1.83669 cm^{-1}, respectively. 2
   (e) The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the moment of inertia (I) and bond length of CO. The relative atomic weight of C = 12.00 and O = 15.9994, the absolute mass of $H = 1.67343\times10^{-27}$ kg. 4

3. (a) How many normal modes of vibration are possible for the following molecules: HBr, O_2, OCS (linear), and SO_2 (bent) 2
   (b) Write the equation for the vibrational energy of a simple harmonic oscillator (such as a diatomic molecule). Using this equation, demonstrate that the diatomic molecule can never have zero vibrational energy. What is the selection rule for a harmonic oscillator undergoing vibrational transitions? 3
   (c) Explain why spacing between vibrational levels in a Morse curve decreases gradually? What are the zero point energies in relation to harmonic and anharmonic oscillators? 2+1
   (d) Sketch the symmetry and fundamental vibrations of the carbon dioxide molecule in a figure. Explain them clearly. 2
   (e) Explain why the C=O stretching vibration of an aldehyde gives rise to a strong absorption in the infrared, although the absorption due to C=C vibration in an alkene is normally very weak. 2
   (f) The equilibrium oscillation frequency for NO is 1904 cm^{-1} when examined in the IR region. Calculate the force constant of the molecule. 2

4. (a) Using quantum theory, explain why Stokes lines are more intense than Anti-Stokes lines. 3
   (b) Explain how the symmetric, bending and asymmetric stretching vibrations of CO_2 appear in Raman spectra. 4
   (c) Explain why homonuclear diatomic molecules (like H_2, N_2) are Raman active but IR inactive using classical theory. 4
   (d) Define the mutual exclusion principle and illustrate its application in distinguishing between CO_2 and CS_2. 3

5. (a) What is the energy of an atomic orbital? What electronic transitions give rise to the Lyman, Balmer, and Paschen series? 4

(b) Given the term symbols $^2P_{3/2}$, $^4D_{5/2}$, and $^3F_4$, find the values of L, S, and J, and explain their physical significance. 3

(c) Draw the energy level diagram for a two-electron atom and indicate possible transitions for paired and unpaired spins. Explain the difference in their spectra. 4

(d) Explain the concept of fine structure in atomic spectra. What causes the splitting of spectral lines in hydrogen? 3

6. (a) Given an absorption spectrum with several vibrational bands, identify the $0\rightarrow0$, $0\rightarrow1$, $0\rightarrow2$ transitions and describe the vibrational progression. 3

(b) State the Franck–Condon principle. Explain, with suitable diagrams, how it governs electronic transitions in a diatomic molecule. 4

(c) Write down the spectral regions for the following electronic transitions and give one example molecule for each: 4
i) $\sigma \rightarrow \sigma^*$; ii) $n \rightarrow \sigma^*$; iii) $\pi \rightarrow \pi^*$; iv) $n \rightarrow \pi^*$

(d) Explain the difference between fluorescence and phosphorescence with diagrams. 3

7 (a) What type of nuclei can exhibit NMR spectrum? Give examples. 1

(b) Deduce an expression to determine the extent of interaction between spin and magnetic field. 3

(c) Define Larmor motion. Deduce an equation for the Lamor frequency of a proton in a magnetic field. 3

(d) What is meant by coupling? How coupling splits NMR absorption line? 3

(e) What is meant by chemical shift? How could you distinguish between *cis*- and *trans*-isomer from ^{1}H-NMR spectra? 1+1

(f) Discuss the anisotropic effect on the chemical shift values. 2

8 Write short notes of **any four** of the following: 3.5×4

(a) Born-Oppenheimer approximations

(b) P, Q, and R branches in the vibration-rotation spectra

(c) Mechanism of spin-spin splitting

(d) Classical theory of Raman effects

(e) Circular dichroism (CD) spectroscopy

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2023
Course Title: Chemical Spectroscopy
Course: Chem. 340 Full Mark: 70 Time: 4 hours

(Answer any five questions)

1(a) What is electromagnetic radiation? Mention its important characteristics. 3
(b) Arrange different types of electromagnetic radiation with increasing order of energy. 3
(c) Why does an atom create a line spectrum whereas a molecule produces a band spectrum? 2
(d) Discuss important factors that contribute to intensity of spectral lines. 3
(e) Calculate the energy associated with 1.00 mol of photons of green light (555 nm) [$c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ Js]. 3

2(a) Which of the following molecules will show a microwave rotational spectrum? H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6? 2
(b) Classify the following molecules on the basis of the principle moment of inertia: HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH. 2
(c) Write down the energy expression in cm^{-1} for a diatomic molecule based on the rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. 3
(d) What are the limitations of the rigid rotator model? Write down the energy expression for the non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra. 4
(e) Calculate the moment of inertia of the water molecule around the axis defined by the bisector of the HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg). 3

3(a) Compare the energy of an anharmonic oscillator and a harmonic oscillator. Calculate the absorption energy of the fundamental band, first overtone, second overtone, and hot band for an anharmonic oscillator. 3
(b) According to Born—Oppenheimer approximation, derive an expression for a diatomic Vibrating-rotator that undergoes υ=0 to υ=1 transition. 4
(c) How many fundamental modes of vibration are possible for H_2O molecules? Are all the vibration modes IR active? Explain with reason. 4
(d) How do you differentiate between CO_2 and N_2O molecules by vibrational spectroscopy, although both are linear triatomic molecules? 3

4(a) Explain the classical theory of Raman effect with example. 3
(b) Why all the vibration modes of CO_2 molecule is not Raman inactive althougn it change the polarizability ellipsoid. Justify your statement. 4

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(c) How do you obtain pure rotational Raman spectra? When we get S branch and show stoke and anti-Stokes' lines for S branch. 4

(d) What is meant by the law of mutual exclusion? How do you distinguish between H_2O and HCN using Raman and infrared spectroscopy? 3

5(a) Show the electronic energy level and possible transition for an electron without and with *J*-splitting. What are the reason for *J*-splitting? 4

(b) Write the term symbol for the electrons with a total spin angular momentum of 2, 1, 3/2 and orbital angular momentum of 0, 1, 2 respectively. 3

(c) Which transition of atomic orbital give rises to the compound doublet and triplet spectra? 3

(d) Explain the energy level and the transition of the He atom whose electrons are in a triplet state. 4

6(a) State Franck—Condon principle. Explain the different electronic transitions for a diatomic molecule according to the Franck—Condon principle. 4

(b) What type of electronic transition occurs in the molecule having the following bonds; (i) C—C, (ii) O—H, (iii) C=N. Mention the spectral region of these transitions. 3

(c) Which of the following molecules give rise to the electronic spectra in higher wavelength? Explain why? (i) CH_4 (ii) CH_3NH_2 (iii) $H_2C=CH-CH=CH_2$ 3

(d) Write the name and explain graphically the different ways by which an electronically excited molecule can lose energy. 4

7(a) Deduce an expression to determine the extent of interaction between spin and magnetic field. What is Larmor frequency? 2.5+1

(b) What is meant by chemical shift? How could you distinguish between *cis*- and *trans*-isomer from ^{1}H-NMR spectra? 1+1

(c) Discuss the anisotropic effect on the chemical shift values. 3

(d) Explain why TMS is used as a reference substance in NMR. 1.5

(e) What is ESR? Explain hyperfine structure with an example. 4

8 Write short notes on any four of the following: 3.5×4

(a) Signal-to-noise ratio

(b) The stark effect and its application

(c) Principle of Mössbauer spectroscopy

(d) Quantum theory of the Raman effect

(e) ESR spectroscopy

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2022
Course Title: Chemical Spectroscopy
Course No.: Chem. 340, Full Marks: 70, Time: 4 hours

(Answer *any five* questions)

1. (a) What is meant by electromagnetic radiation? Write the electromagnetic radiation that has the highest and the lowest wavelength. Mention its important properties. 4.0
   (b) Explain the term 'Quantization of Energy' with an example. 3.0
   (c) What is meant by signal-to-noise ratio? How do you resolve it? 4.0
   (d) Why a molecule does produce a band spectrum whereas an atom produces a line of the spectrum? 3.0

2. (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 2.0
   (b) Mention the limitations of microwave spectroscopy. 2.0
   (c) In the rotational spectrum of CO at 25^0 C, the $J = 3 \rightarrow J = 4$ transition is more intense than the $J = 0 \rightarrow J = 1$ transition. Explain. 3.0
   (d) How do you classify the molecules according to the rotation along the three directions? Give example. 3.0
   (e) The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the moment of inertia (I) and bond length of CO. The relative atomic weight C= 12.00 and O = 15.9994, the absolute mass of H= 1.67343×10^{-27} kg. 4.0

3. (a) How many normal modes of vibration are possible for the following molecules: (i) SO_2 (bent), (ii) BCl_3, (iii) $HC{\equiv}CH$, (iv) CH_4. 2.0
   (b) Write and compare the energy of the anharmonic oscillator and harmonic oscillator. Using the Morse curve, mention the allowed vibrational energy levels and some transitions for a diatomic molecule undergoing anharmonic oscillations. Write the name of the band observed by these transitions. 4.0
   (c) Why does the linear CO_2 exhibit different vibration modes than expected? Compare with the similar triatomic H_2O. 4.0
   (d) Write the energy of a diatomic vibrating-rotator. How do you observe the O, P, Q, R, and S bands for this molecule? 4.0

4. (a) According to the quantum theory of Raman spectra explain the stokes and anti-stokes radiation. Which radiation has a higher frequency? 3.0
   (b) Explain the Raman activity of the fundamental modes of vibration in the H_2O molecule. 3.0
   (c) Symmetric top molecules are Raman active but spherical top molecules are Raman inactive. Explain with examples. 4.0
   (d) What is the rule of mutual exclusion? How do you predict the Raman and infrared active bands as well as shape for the following triatomic molecules? 4.0
   i) CO_2, ii) N_2O, iii) SO_2

5. (a) Explain the energy of an atomic orbital for the hydrogen atom. Which electronic transition can give rise to the Lyman and Balmer spectral lines? Mention their absorption energy. 3.0
   (b) Find the value of L, S, and J for $^4P_{3/2}$ and $^2D_{5/2}$. What do these values signify? 3.0

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(c) What are the factors that govern the fine structure of the hydrogen atom spectrum? 2.0

(d) What are the selection rules for many-electron systems in the case of electronic spectroscopy? 3.0

(e) Which transition of electrons gives rise to the doublet and compound doublet state for Na atom? Write the selection rule for this transition. 3.0

6. (a) State Franck—Condon principle. Discuss the different possibilities of transitions for a molecule which undergoes transition from ground state to upper state both electronically and vibrationally. 4.0

(b) Explain why homonuclear molecules such as H_2 and N_2 do not show vibrational and rotational spectra but they exhibit electronic spectra. 3.0

(c) What are the different processes by which excited molecules can loss its excess energy? Discuss the fluorescence process with necessary diagram. 1+2=3

(d) Discuss Wood-Fieser's rule to calculate the wavelength of maximum absorption with example. 2.0

(e) Sketch and discuss the UV-Visible spectrum of propanone and benzene. 2.0

7. (a) What type of nuclei can exhibit NMR spectrum? Give examples. 1.0

(b) Deduce an expression to determine the extent of interaction between spin and magnetic field. 3.0

(c) What is meant by chemical shift? Explain why TMS is used as a reference substance in NMR. 3.0

(d) Differentiate between fine structure and hyper fine structure of ESR absorption. 3.0

(e) How could you distinguish between *cis*- and *trans*- isomer from ^{1}H-NMR spectra? 2.0

(f) Describe the anisotropic effect on the chemical shift value. 2.0

8. Write short notes of *any four* of the following: 4+4+3+3 = 14

(a) The intensity of spectral lines

(b) The stark effect and its application

(c) ESR spectroscopy

(d) The classical theory of the Raman effect

(e) Principle of Mössbauer spectroscopy

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2021
Course Title: Chemical Spectroscopy
Course: Chem. 340 Full Mark: 70 Time: 4 hours

(*Answer any five questions*)

1. (a) What is electromagnetic radiation? Mention its important characteristics. 3
   (b) Mention the name of different regions of spectrum according to the increasing frequency of the regions. 2
   (c) What is meant by quantization of energy? Illustrate how all forms of energies are quantized. 1+2= 3
   (d) Draw and label a block diagram of single beam spectrophotometer and describe each component mentioning its function. 3
   (e) The wavelength associated with a photon is 500 nm. Calculate its wavenumber, frequency and energy [Given that, $c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ Js]. 3

2. (a) Which of the following molecules will show a microwave rotational spectrum: H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6? 2
   (b) Classify the following molecules on the basis of the principle moment of inertia: HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH. 2
   (c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. 3
   (d) What are the limitations of the rigid rotator model? Write down the energy expression for non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra. 4
   (e) Calculate the moment of inertia of water molecule around the axis defined by bisector of HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg). 3

3. (a) Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic and an anharmonic oscillators. Show the allowed transitions and the expected spectrum. 3
   (b) How vibrational spectrum of a diatomic molecule enables us to determine anharmonicity constant and equilibrium frequency of vibration? 3
   (c) What are P, Q and R branches in the vibration-rotation spectra? Explain their origin. 3
   (d) How can intermolecular and intramolecular hydrogen bonding be identified by IR spectroscopy? 2
   (e) The fundamental band of CO in inert carbon tetrachloride solvent is found at 2143 cm^{-1} when examined in the IR region. Calculate the force constant of the molecule. 3

4. (a) What are meant by stokes and anti-stokes radiation? 2
   (b) Define molecular polarizability. State the criteria for Raman activity with example. 3
   (c) Explain how all stretching and bending vibrational modes of water are obtained to be Raman active. 3
   (d) State and explain the rule of mutual exclusion with example. 3
   (e) State three differences between Raman technique and Infra-Red vibrational measurements. 3

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5. (a) Describe Born-Oppenheimer approximation for electronic spectra of diatomic molecules. Explain why all molecules can exhibit electronic spectra. 2+1= 3
(b) Explain how the dissociation energy can be determined by the analysis of the electronic spectra of a diatomic molecule. 4
(c) Write down the spectral region in which the following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$. 2
(d) What are the different processes by which an excited molecule may loss its excess energy? Discuss phosphorescence process with energy diagram. 1+2.5=3.5
(e) What are the basic differences between electronic and Raman spectroscopy? 1.5

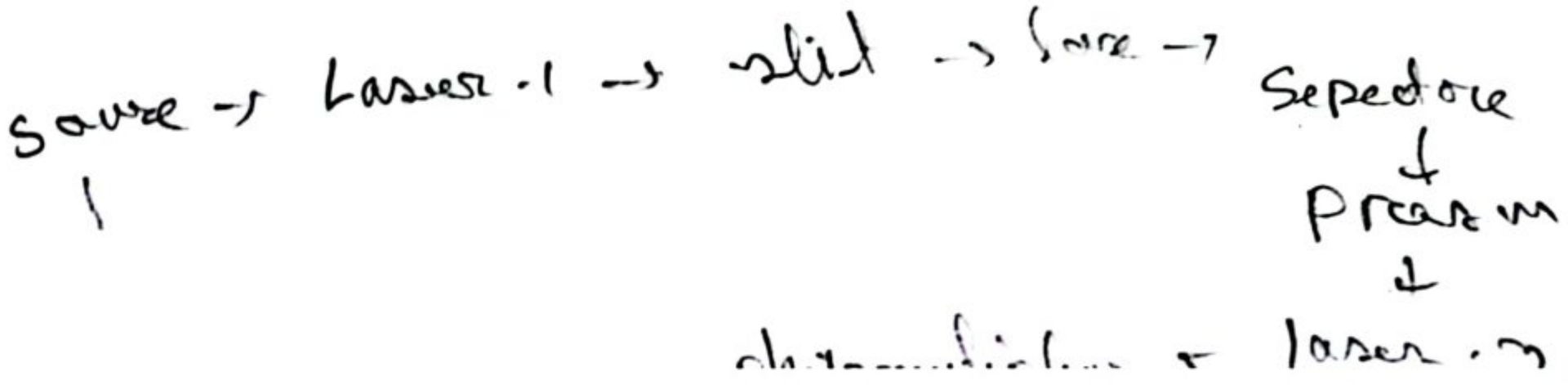
6. (a) State and explain building up principles with example. 3
(b) What are meant by hydrogen- and helium-like species? Give examples. 2
(c) Show with spectrum that the lines of Lymen series of hydrogen atom are observed at 3R/4, 8R/9, 15R/16, ------ cm^{-1} in the spectrum.
(d) Describe the transitions from ^2P to ^2D states for hydrogen atom spectrum with necessary energy diagram and spectrum. Why is the spectrum obtained from $^2P \rightarrow {}^2D$ transitions referred to as compound doublet? 3+3
(e) Find the values of L, S and J for $^4S_1$ and $^2D_{7/2}$.

7. (a) What is meant by Larmor frequency? Show that the Larmor frequency of a spinning nucleus in a magnetic field is just equal to the frequency separation between energy levels. 3
(b) If a specific NMR instrument operates at 400 MHz, calculate the magnetic field strength which is required for ^1H nucleus to resonance at this frequency. Given that plank constant (h) is 6.62×10^{-34} J s^{-1} and $\beta_N = 5.05 \times 10^{-27}$ JT^{-1}. 3
(c) Why is TMS almost universally used as a standard reference in NMR spectroscopy? Which solvents can be used in the preparation of samples in NMR spectroscopy? 1.5+0.5=2
(d) Explain why it is more convenient to refer ESR absorption in terms of its 'g' factor. Why is the 'g' factor value varied to a large extent in case of ionic crystals of transition metal ions? 2
(e) State the criteria for different species to exhibit the ESR spectroscopy. Discuss ESR spectrum of benzene radical. 2+2=4

3.5 × 4 = 14

8. Write short notes of any four of the following:
(a) Frank-Condon principle
(b) Classical theory of Raman effects
(c) Spectrum of Helium atom
(d) Circular dichroism (CD) spectroscopy
(e) Mössbauer spectroscopy

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340

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Chemical Spectroscopy
Course # Chem. 340, Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1 (a) What are meant by induced absorption and stimulated emission processes? Illustrate these processes with energy diagram and example. 1.5+1.5
(b) State the characteristics of electromagnetic radiation. Write the different regions of spectrum according to the increasing frequency of the regions. 1.5+0.5
(c) Illustrate the different factors which influence the intensity of spectral lines. 3.0
(d) Sketch a diagram of a grating spectrometer. State the purposes of the use of slits in grating spectrometer. 1+1
(e) What is meant by Doppler broadening? How could you remove Doppler broadening? 4.0

2 (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 1+2
(b) Mention the limitations of microwave spectroscopy. 2.0
(c) In the rotational spectrum of CO at 25° C, the $J = 3 \rightarrow J = 4$ transition is more intense than the $J = 0 \rightarrow J = 1$ transition. Explain. 3.0
(d) Explain the reason why the rotational constants of $^{12}C^{16}O$ and $^{13}C^{16}O$ are 1.92118 cm^{-1} and 1.83669 cm^{-1}, respectively. 2.0
(e) The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the moment of inertia (I) and bond length of CO. The relative atomic weight C = 12.00 and O = 15.9994, the absolute mass of H = 1.67343×10^{-27} kg. 4.0

3 (a) How many normal modes of vibration are possible for the following molecules: HBr, O_2, CH_4 and BCl_3 2.0
(b) Write down the equation for the vibrational energy of a simple harmonic oscillator (a harmonic diatomic molecule). From this equation show that the diatomic molecule can never have zero vibrational energy. What is the selection rule for harmonic oscillator undergoing vibrational changes? 3.0
(c) Explain why does spacing between vibrational levels in a Morse curve decrease gradually? What are the zero point energies in relation to harmonic and anharmonic oscillators? 2+1
(d) Sketch the symmetry and fundamental vibrations of the carbon dioxide molecule in a figure. Explain them clearly. 2.0
(e) Explain why the C=O stretching vibration of an aldehyde gives rise to a strong absorptionin the infrared, although the absorption due to C=C vibration in an alkene is normally very weak. 2.0
(f) The equilibrium oscillation frequency for NO is 1904 cm^{-1} when examined in the IR region. Calculate the force constant of the molecule. 2.0

4. (a) State and explain the building up principle for many-electron atoms withy example. 3.0
(b) (i) Sketch the electronic energy levels and show transition between them for single electron of hydrogen atom. 1.5
(ii) State the selection rule for H-atom spectrum. Which of the following transitions are allowed for H-atom spectrum? 1.5
$2p \rightarrow 4d$, $3p \rightarrow 4s$, $2p \rightarrow 4f$, $2p \rightarrow 1s$

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(iii) Sketch the lines of Lyman series observed at $\frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16}, \frac{24R}{25}, \ldots$ cm^{-1} in the spectrum of H-atom. 1.5

(c) What are the values of L, S and J for the term symbols $^4P_{3/2}$ and $^2D_{5/2}$. 2.5

(d) Discuss the transitions between ^3P and ^3D levels in the helium atom with necessary energy diagram and spectrum. Why the spectrum obtained from $^3P \rightarrow {}^3D$ transitions is referred to as compound triplet. 3+1

5. (a) State Frank-Condon principle. Discuss the different possibilities of transition for a molecule which undergo electronic transition from ground state to upper state both electronically and vibrationally. 4.0

(b) Explain why homonuclear molecules such as H_2 and N_2 do not show vibrational and rotational spectra but they exhibit electronic spectra. 3.0

(c) Write down the spectral region in which the following transitions may take place : 2.0

$$\sigma \rightarrow \sigma^*, n \rightarrow \sigma^*, \pi \rightarrow \pi^*, n \rightarrow \pi^*.$$

(d) What are the different processes by which an excited molecule may loss its excess energy? Discuss phosphorescence process with energy diagram. 1+2.5

(e) What are the basic differences between electronic and Raman spectroscopy? 1.5

6. (a) Deduce an expression to determine the extent of interaction between spin and magnetic field. What is Larmor frequency? 2.5+1

(b) What is meant by chemical shift? How could you distinguish between *cis*-and *trans*-isomer from ^{1}H-NMR spectra? 1+1

(c) Discuss the anisotropic effect on the chemical shift values. 3

(d) Explain why TMS is used as reference substance in NMR. 1.5

(e) What is ESR? Explain hyperfine structure with example. 4.0

7. (a) Illustrate the quantum theory of Raman effect. Mention the differences between stokes and anti-stokes radiation. 2.5+1.5

(b) What are the criteria for a molecule to be Raman active? Give examples of three molecules which exhibit Raman spectra. 2.0

(c) Write energy expression and selection rule of Raman spectra in case of symmetric top molecules. Also show that *R* branch lines appear at 4B, 6B, 8B, 10B ... cm^{-1} from the exciting line. 3.0

(d) State and explain the rule of mutual exclusion with example. 3.0

(e) Discuss the advantages of Raman technique over that of Infra-Red in vibrational measurements. 2.0

8. Write informatory notes on any four of the following : 3.5×4

(a) Anisotropic effect on Chemical shift

(b) Mossbauer Spectroscopy

(c) The positions of E.S.R. Absorptions: the g factor

(d) Quantization of Energy

(e) Building-up principles

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2018
Course Title: Chemical Spectroscopy
Course No. Chem. 340; Full Mark: 70
Time: 4 hours

(*Answer any five questions*)

1. (a) What is electromagnetic radiation? Mention its important characteristics. 3
   (b) Arrange different types of electromagnetic radiation with increasing order of energy. 2
   (c) Why does a molecule produce a band spectrum but an atom a line spectrum? 3
   (d) Discuss important factors that contribute to intensity of spectral lines.
   (e) Calculate the energy associated with 1.00 mol of photons of green light (555 nm) [$c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ Js]. 3

2. (a) Which of the following molecules will show a microwave rotational spectrum? 2
   H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6
   (b) Classify the following molecules on the basis of the principle moment of inertia: 2
   HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH
   (c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. 3
   (d) What are the limitations of the rigid rotator model? Write down the energy expression for non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra. 4
   (e) Calculate the moment of inertia of water molecule around the axis defined by bisector of HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg). 3

3. (a) Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic and an anharmonic oscillators. Show the allowed transitions and the expected spectrum. 3
   (b) How vibrational spectrum of a diatomic molecule enables us to determine anharmonicity constant and equilibrium frequency of vibration? 3
   (c) What are *P*, *Q* and *R* branches in the vibration-rotation spectra? Explain their origin. 3
   (d) How can intermolecular and intramolecular hydrogen bonding be identified by IR spectroscopy? 2
   (e) The fundamental band of CO in inert carbon tetrachloride solvent is found at 2143 cm–1 when examined in the IR region. Calculate the force constant of the molecule. 3

4. (a) Explain the quantum theory of Raman effect. What are the differences between stokes and anti-stokes radiation? 2.5+1=3.5
   (b) State the criteria for a molecule to be Raman active. Give at least three examples of molecules which can give Raman spectra. 2.5
   (c) Symmetric stretching mode of CO_2 molecule is Raman active whereas bending and asymmetric stretching modes are observed to be Raman inactive. Explain the reasons with the plot of the variation of polarizability with displacements during vibration. 3
   (d) State and explain the rule of mutual exclusion with example. 2
   (e) Explain why the pure rotations of spherical top molecules are completely inactive in Raman spectroscopy. 1
   (f) State the advantages of Raman technique over that of Infra-Red in vibrational measurements. 2

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5. (a) Write down the wave function expression and energy expression for H atom.
   (b) What are atomic quantum numbers? Discuss their significance.
   (c) What are the values of L and S for the term symbols $^3S_1$ and $^2P_{3/2}$.
   (d) Explain compound doublet spectra with energy diagram and spectrum for the transition between ^2P and ^2D levels in the fine H-atom spectra. 3
   (e) Sketch the energy levels of lithium atom in a figure showing the difference in energy of *s*, *p*, and *d* states with the same value of *n* and also show some allowed transitions in the figure. 3

6. (a) State Franc-Condon Principle. Discuss the different possibilities of transition for a molecule which undergoes electronic transition from ground state to upper state both electronically and vibrationally. 4
   (b) Explain why homonuclear molucules such as H_2 and N_2 do not show vibrational and rotational spectra but they exhibit electronic spectra. 3
   (c) What are the different processes by which excited molecules can loss its excess energy? Discuss the fluorescence process with necessary diagram. 1+2.5 = 3.5
   (d) Write down the spectral region in which the following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$. 2
   (e) What are the basic differences between electronic and Raman spectroscopy? 1.5

7. (a) Which of the following nuclei can exhibit NMR spectrum:
   ^{37}Cl, ^{12}C, ^{1}H, ^{13}C, ^{19}F, ^{16}O and ^{31}P? 2
   (b) Describe an expression to determine the extent of interaction between spin and magnetic field. 3
   (c) If a particular NMR instrument operates at 300.26MHz, calculate the magnetic field strength which is required for ^1H nucleus to resonance at this frequency. Given that plank constant (h) is $6.62 \times 10\text{-}34$ J s^{-1} and $\beta_N = 5.05 \times 10\text{-}27$ JT^{-1}. 3
   (d) Why is TMS almost universally used as a standard reference compound for hydrogen resonances in NMR spectroscopy? 2
   (e) What is meant by chemical shift? How could you differentiate trans- and cis- isomers with the concept of chemical shift? 2
   (f) What are the criteria for different species to be active in ESR spectroscopy? Discuss with examples of molecules/ions/radicals which can exhibit ESR spectrum. 2

8. Write short notes of <u>any four</u> of the following: 3.5 × 4 = 14
   (a) Width and intensity of spectral transitions
   (b) Born-Oppenheimer approximation
   (c) Chemical analysis by Raman spectroscopy
   (d) Spectrum of Helium atom
   (e) Circular dichroism (CD) spectroscopy
   (f) Mössbauer spectroscopy

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$m^{-2} \times kg \times m^2 s^{-2}$

$kg\, s^{-2}$

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) (Part III) Final Examination 2017
Course Title: Chemical Spectroscopy
Course No. Chem. 340F
Time: 4 Hours, Total Marks: 70

[Answer any five questions]

1. (a) What are the differences between induced absorption and stimulated emission processes? 2

(b) Mention the characteristics of electromagnetic radiation. Mention the name of different regions of spectrum according to the increasing frequency of the regions. 2

(c) Sketch a diagram of a grating spectrometer. Mention the purposes of the use of slits in grating spectrometer. 2.5

(d) What is meant by quantization of energy? Illustrate how all forms of energies are quantized. 2.5

(e) State the factors which cause the broadening of spectral lines. Explain how molecular effusive beam can be produced. Describe effusive molecular beam method to remove pressure broadening. 2+1+2

2. (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 2

(b) Mention the limitations of microwave spectroscopy. 1.5

(c) Describe briefly the effect of different factors on the rotational energy and spectrum. 3

(d) Discuss the mechanism of action of a microwave oven. What are the differences between microwave oven and conventional oven? 2+1

(e) Write the energy expression and show allowed transitions between rotational energy levels for rigid diatomic molecule. Also show that absorption spectrum of rigid diatomic molecule consists of lines at 2B, 4B, 6B, 8B, ----- cm^{-1}. 1.5+1.5

(f) The first line of carbon monoxide molecule for the transition from J=0 to J=1 was observed to be as 3.84235 cm^{-1}. Calculate the rotational constant (B) and moment of inertia (I). 1.5

3. (a) How many normal modes of vibration are possible for the following molecules: HBr, O_2, OCS (linear) and SO_2 (bent) 2

(b) Write down the equation for the vibrational energy of a simple harmonic oscillator (a harmonic diatomic molecule). From this equation show that the diatomic molecule can never have zero vibrational energy. What is the selection rule for harmonic oscillator undergoing vibrational changes? 1+1+0.75

(c) Explain why spacing between vibrational levels in a Morse curve decreases gradually? What are the zero point energies in relation to harmonic and anharmonic oscillators? 2+1

(d) Sketch the symmetry and fundamental vibrations of the carbon dioxide molecule in a figure. Explain them clearly. 2.25

(e) Explain why the C=O stretching vibration of an aldehyde gives rise to a strong absorption in the infrared, although the absorption due to C=C vibration in an alkene is normally very weak. 1.5

(f) The equilibrium oscillation frequency for NO is 1904 cm^{-1}. Calculate the force constant of the molecule. 2.5

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4. (a) What is meant by 'Raman scattering'? Mention the criteria of a molecule that makes it Raman active? 1+1.5

(b) (i) Write down the energy expressions for pure rotational Raman spectra for a linear molecule. 1

(ii) Show the rotational energy levels of a diatomic molecule and the rotational Raman spectrum arising from them with showing the relative intensities of lines in the spectrum. 1.5

(iii) Explain the Stokes' and anti-Stokes' lines in the rotational Raman spectrum of the above figure (ii). 1.5

(c) What is the rule of mutual exclusion? What structural information can you get from Raman spectral analysis? 1+2

(d) With which type of spectroscopy would one observe the pure rotation spectrum of H_2? If the bond length of H_2 is 0.07417 nm, what would be the spacing of the lines in the spectrum? 0.5+2

(e) "CO_2 shows two bands in IR spectrum but one in Raman spectra" – explain the term. 2

5. (a) Describe Born-Oppenheimer approximation for electronic spectra of diatomic molecules. Explain why all molecules can exhibit electronic spectra. 1.5+1

(b) Illustrate progressions with allowed electronic transitions and spectrum. 3.5

(c) Write the name of different processes by which excited molecules can loss its excess energy. Discuss the phosphorescence process with necessary diagram. 1+2

(d) Differentiate between Fluorescence process and Raman spectroscopy? 2

(e) Sketch and discuss the UV-Visible spectrum of propanone and benzene. 3

6. (a) State and explain building up principle for many electron atoms with example. 3

(b) (i) Sketch the electronic energy levels and show transition between them for single electron of hydrogen atom. 1+1.5+1.5

(ii) State the selection rule for H-atom spectrum. Which of the following transitions are allowed for H-atom spectrum?

$2p \rightarrow 4d$, $3p \rightarrow 4s$, $2p \rightarrow 4f$ and $2p \rightarrow 1s$

(iii) Show with spectrum that the lines of Lyman series are observed at 3R/4, 8R/9, 15R/16, ---- cm^{-1} in the H-atom spectrum.

(c) (i) What is the selection rule for many electron systems in case of electronic spectroscopy? 0.5+3

(ii) Discuss the transitions from ^3P to ^3D states for Helium atom spectrum with necessary energy diagram and spectrum. Why the spectrum obtained from $^3P \rightarrow {}^3D$ transitions is referred to as compound triplet?

(d) Find the values of L, S and J for $^4P_{3/2}$ and $^2D_{5/2}$.

(e) What are meant by hydrogen- and helium-like species? Give examples.

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2016
Course Title: Chemical Spectroscopy
Course No: Chem - 340
Time: 4 Hours, Total Marks: 70

Answer any five questions

1. (a) What is meant by induced absorption? Differentiate between spontaneous emission and stimulated emission.. 1+1.5
   (b) State the characteristics of electromagnetic radiation. 1.5
   (c) What is your concept of quantization? Explain why any form of energies of a molecule is quantized. 1+2.5
   (d) How could you remove Doppler line broadening with the help of Lamb dip spectroscopic method? 4
   (e) Sketch the spectra for the following: 3
   (i) ^{1}H-NMR spectrum of ethanol and 2
   (ii) Ultra-violet spectrum of propanone. 2
   (f) What are the purposes of the use of slits in grating spectrometer? 2

2. (a) Which of the following molecules will show a microwave rotational spectrum: 2
   H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6
   (b) Classify the following molecules on the basis of the principle moment of intertia. 2
   HCl, CH_4, CH_3F, H_2O, OCS, BCl_3
   (c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator. Show that the absorption spectrum of the rigid diatomic molecule is expected to consist of a series of equidistant lines. 3
   (d) (i) Write down the energy expression in cm^{-1} for a non-rigid rotator. Compare the change in rotational energy levels and spectrum when passing from a rigid to a non-rigid diatomic molecule with the help of a figure. 0.5+2
   (ii) explain the effect of isotopic substitution on rotational spectra of a diatomic molecule such as coarbon monoxide with the help of appropriate figures and equations. 1.5
   (e) The rotational spectrum of $^{79}B^{19}F$ shows a series of equidistant lines 0.71433 cm^{-1} apart. Calculate the rotational constant B, and hence the moment of inertia and bond length of the molecule.. 3

3. (a) State the basic requirement for a substance to be IR active with example. 1
   (b) Sketch the Morse curve for the anharmonic oscillator. Give the selection rule for anharmonic oscillator. Explain fundamental, hot band, and overtone transitions with energy expressions. 1+1+3
   (c) The fundamental frequency of vibration for $H^{35}Cl$ molecule is 2886 cm^{-1}. Calculate the force constant of the $H^{35}Cl$ molecule. Given that atomic mass of ^{35}Cl = 34.97 amu. 3
   (d) Explain why all transitions in vibrational spectroscopy are restricted to originate from ground state (υ=0). 2
   (e) What are P and R branch lines? Sketch vibrational-rotational energy levels and show the transitions between them to appear P and R branch lines in vibration-rotation spectrum. 3

4. (a) Which modes of vibration in H_2O molecule are Raman active and why? 2
   (b) Consider the symmetric top molecule $CHCl_3$
   (i) Write down the equation for the energy levels of the molecule. 1
   (ii) What are the selection rules for Raman spectra in this case? 1
   (iii) write the expression for the wavenumber ($\bar{\nu}_k$ and $\bar{\nu}_s$) equations with mentioning

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their J and ΔJ values at which two series of R branch and S branch lines appear in the Raman spectrum. 2

(iv) Sketch the S branch series only in a Figure and show that the S branch series occur at $6B$, $10B$, $14B$, ---------- cm^{-1} from the exciting line. 3

(c) The wavenumber of the incident radiation in a Raman spectrometer is 20487 cm^{-1}. What is the wave number of the scattered Stokes radiation for the $J = 2 \leftarrow 0$ transition of $^{14}N_2$? [Given that $B = 1.9987\ cm^{-1}$]. 3

(d) Which of the vibrations of an AB_3 molecule are infrared or Raman active when it is (i) trigonal planar (symmetric planar), (ii) trigonal pyramidal (Symmetric pyramidal)? 2

5. (a) Get the values for L and S, for $^3S_1$, $^2P_{3/2}$. 1.5

(b) How can you get the term symbol for the following? 1+1

(i) $L = 1, S = 1$ and $J = 2$

(ii) $L = 2, S = ½$ and $J = 3/2$ 1.5

(c) What is meant by the 'j-splitting'?

(d) (i) Sketch the energy levels of lithium atom in a figure showing the difference in energy of s, p, and d states with the same value of n and also show some allowed transitions in the figure. 2

(ii) The doublet separation of lines in the spectral series, which is scarcely observable for hydrogen, is less than 1 cm^{-1} for the $2p$ level of lithium, about 17 cm^{-1} for sodium and over 5000 cm^{-1} for cesium' –explain the term. 1

(e) Explain compound doublet spectra with energy diagram and spectrum for the transition between ^2P and ^2D levels in the fine H-atom spectra. 3

(f) Why He^+, Be^+, B^{2+} etc., show 'hydrogen-like spectra'? 1

(g) Explain why the lines in atomic spectra of Na are doulet. 2

6. (a) What is the Born-Oppenheimer approximation of electronic spectra of diatomic molecules? Explain why all molecules can give electronic spectra. 2+1

(b) Discuss Woodward-Feiser's rule to calculate the wavelength of maximum absorption with example. 2

(c) State Franc-Condon principle. If an electronic transition takes place for a molecule at ground state both electronically and vibrationally to upper state, discuss the different possibilities of transition of that molecule. 1+3

(d) What are the different processes by which an excited molecule may loss its excess energy? Discuss phosphorescence process with energy diagram. 1+2.5

(e) What are the basic differences between electronic and Raman spectroscopy? 1.5

7. (a) Deduce the expression to determine the extent of interaction between spin and magnetic field. What is Larmor frequency? 2.5+1

(b) What is meant by chemical shift? How could you differenciate between *cis*- and *trans*-isomer form ^1H-NMR spectra? 1+1

(c) Discuss the anisotropic effect on the chemical sift values. 3

(d) Explain why TMS is used as reference substance in NMR. 1.5

(e) What type of substance can exhibit E.S.R. spectrum? Give examples. 2

(f) Sketch the E.S.R. spectrum of benzene radical. Find the total electron density for six hydrogens of this radical. 2

8. Write informatory notes on any four of the following: $3.5 \times 4 = 14$

(a) Width and intensity of spectral transitions

(b) Breakdown of Born-Oppenheimer approximation

(c) Spectrum of Helium atom

(d) Classical theory of Raman effects, and

(e) The positions of E.S.R absorptions.

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